

Data Structure and Algorithms



Objectives



- Data Structures
- Algorithms



Course Description

- Covers design, analysis and implementation of basic data structures
- Algorithms operating on data structures are also covered.
- Asymptotic analysis of algorithms and data structures are also covered.



Course Contents

- Introduction to Data Structures
- Linked Lists
- Stacks & Queues
- Non-linear Data Structures
- Algorithms (Searching & Sorting)
 - [Linear Search, Binary Search], [Bubble sort, Selection sort, insertion sort]
 - [Hashing], [Quick Sort, Heap Sort, Merge Sort]
- Trees & Graphs



Prerequisite and assessment

□ Fundamentals of computer science

□ Assessment:

Mid Exam – 20%

Lab Exam – 20%

Assignment – 10%

Final Exam – 50%



Syllabus Programming Languages

- Either **Java** or **Python** can be used for practice and exam.
- Classes will be conducted using **Java**
- **IDEs: Eclipse** or NetBeans are fine.
- Data Structures and Algorithms are independent of any programming language.
- Data structure and algorithms are general programming concepts and once you are familiar with concepts you can work in any language.



Syllabus Reading

□ We will not have a single book, instead we will use many books, websites, tutorials

□ Books

- Data Structures and Algorithms in **Java**, 2nd edition, Robert Lafore
- Data Structures and Algorithms in **Java**, 6th edition, Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser
- Data Structures and problem Solving in **Java**, 4th edition, Mark Allen Weiss
- Data Structures and Algorithms in **Python**, Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser
- Data Structures and Algorithms Made easy in **Java**, Narasimha Karumanchi



Syllabus -warning

□ If you don't do the labs

- The assignments will be very hard
- The exams will be very hard
- You will not be able to work on your project

□ Copying labs, project and the assignments will result in –ve value 😊



DSA Lab 0 - Installation

□ Installing (For Java users)

- Java (newest JDK preferred)
- Eclipse / NetBeans

□ Installing (For Pythonians)

- VS code or PyCharm or Thonny

□ Running

□ Manipulating Arrays



Introduction to Data Structures

- Discussion points:

Variables

Data Types

OOP Concepts (Object, class, inheritance, abstraction, inheritance, polymorphism, interface)

Data Structures



- **Data structure** is a particular way of storing and organizing data in a computer so that it can be used efficiently.
- It's a special format for organizing and storing data.
- A data structure is an arrangement of data in a computer's memory (or sometimes on a disk).
- **Data structures** include
 - arrays,
 - linked lists,
 - stacks,
 - binary trees, and hash tables, among others.
- **Algorithms** manipulate the data in these structures in various ways, such as searching for a particular data item and sorting the data.



Abstract Data Types(ADTs)

- Abstract Data Type: is an abstraction of a data structure which provides only the **interface** to which a data structure must adhere to.

The interface does not give any specific details about how something should be implemented or in what programming language.

E.g., ADT vs Implementation

List □ Dynamic array or linked list

Queue □ Linked List, Array, Stack

Map □ Tree Map, Hash Map or Hash Table

- Roughly speaking, it's a way of looking at a data structure: focusing on what it does and ignoring how it does its job



What is a Data Structure in Java?

- Data structure refers to a data collection with well-defined operations and behavior or properties.
- A data structure is a unique way of storing or organizing the data in computer memory so that we can use it effectively.
- A data structure is an **arrangement of data in** a computer's **memory** (or sometimes on a disk).

DS types

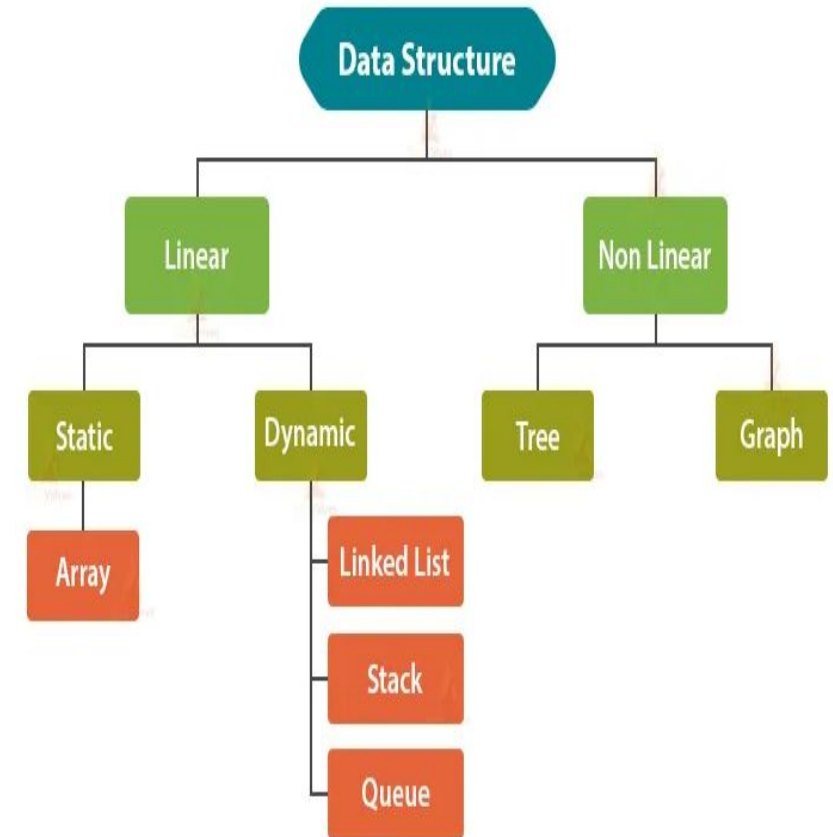
- Based on the organization of elements,

Linear Data Structures –

Elements are accessed in a sequential order

Non-linear Data Structures –

- Elements are accessed in a non-linear order (non-sequential manner)





Algorithms

- a finite sequence of instructions, each of which has a clear meaning and can be performed with a finite amount of effort in a finite length of time
- Engineers use pseudocodes to first draft and describe algorithms
“somewhere between formatted English and computer program code”



Fundamental questions about algorithms

- What is it supposed to do? [specification]
- Does it really do what it is supposed to do? [verification]
- How efficiently does it do it? [performance analysis | efficiency]



- It takes a set of input(s) and produces the desired output.

For example:

An algorithm to add two numbers:

1. Take two number inputs
2. Add numbers using the + operator
3. Display the result



Examples of algorithms

- Searching algorithms
- Sorting algorithms



Types of notations for time complexity

